Chapter 6
The Link between Operations and Finance

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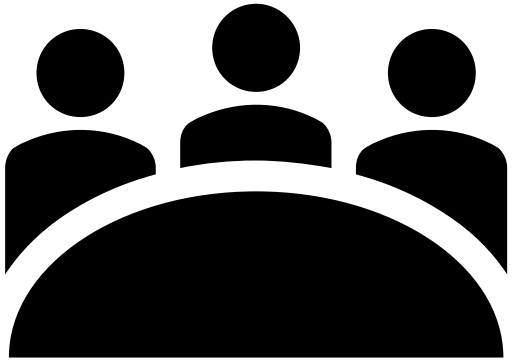
Collaborative Clarification

- Students raise questions about parts they found difficult
- Any students may answer and one additional student may add comments
- The professor clarifies both the questions and answers if necessary
- Students who raised questions and answered write down what they did (A Q&A sheet)



ROIC

- The goal of most companies is to create economic value
  - Economic value means that a firm returned more money than it deposited to the bank
- ROIC (Return On Invested Capital)

$$\text{ROIC} = \frac{\text{Return (Net Income)}}{\text{Invested Capital}}$$

Invested Capital (In the **balance sheet)** = Working capital + PP&E (property, plant and equipment) = (Current assets – Current liabilities) + PP&E

Net Income (In the **income statement)**: A company's total earnings (or profit). Net income is calculated by taking revenues and adjusting for the cost of doing business, depreciation, interest, taxes and other expenses.

Understanding ROIC from Financial Statements

Tesla, Inc. (TSLA)

Balance Sheet All numbers in thousands

Breakdown	12/30/2021	12/30/2020
> Total Assets	62,131,000	52,148,000
> Total Liabilities Net Minority Int...	31,116,000	29,073,000
> Total Equity Gross Minority Inte...	31,015,000	23,075,000
Total Capitalization	34,443,000	30,738,000
Common Stock Equity	30,189,000	22,225,000
Capital Lease Obligations	3,531,000	3,008,000
Net Tangible Assets	28,472,000	21,705,000
Working Capital	7,395,000	12,469,000
Invested Capital	35,531,000	32,496,000
Tangible Book Value	28,472,000	21,705,000
Total Debt	8,873,000	13,279,000
Net Debt	-	-
Share Issued	1,033,000	960,000
Ordinary Shares Number	1,033,000	960,000

Income Statement All numbers in thousands

Breakdown	TTM	12/30/2021
> Total Revenue	53,823,000	53,823,000
Cost of Revenue	40,217,000	40,217,000
Gross Profit	13,606,000	13,606,000
> Operating Expense	7,110,000	7,110,000
Operating Income	6,496,000	6,496,000
> Net Non Operating Interest Inc...	-315,000	-315,000
> Other Income Expense	162,000	162,000
Pretax Income	6,343,000	6,343,000
Tax Provision	699,000	699,000
∨ Net Income Common Stockhold...	5,519,000	5,519,000
> Net Income	5,519,000	5,519,000
Diluted NI Available to Com Stock...	5,519,000	5,519,000

Invested capital =

$$\frac{35531000+32496000}{2}$$

= 34013500

ROIC =

$$\frac{5519000}{34013500} \times 100$$

= 16.2%

Impact of Operational Improvements on ROIC

How can you justify if process improvement contributes to ROIC improvement?

Process improvement should enhance the three key performance measures of an organization
(Flow time, Inventory, & Flow rate)

Hence, let's re-express the ROIC equation w.r.t. the three key performance measures of an organization

Du Pont Analysis

$$\text{ROIC} = \frac{\text{Return (Net Income)}}{\text{Invested Capital}}$$

$$= \frac{\text{Return (Net Income)}}{\text{Revenue (Sales)}} \times \frac{\text{Revenue (Sales)}}{\text{Invested Capital}}$$

Margin Ratio

Capital Turnover
(Capital efficiency)

Remember Little's Law & Inventory Turnover

$$\text{Average FT} = \frac{\text{Average Inventory}}{\text{Average Flow Rate}}$$

$$\text{Inventory Turnover} = \frac{1}{\text{Average FT}}$$

$$= \frac{\text{Average Flow Rate}}{\text{Average Inventory}}$$

Note that Revenue (Sales) is used instead of Cost of Goods

Implications of Du Pont Analysis

- High margin industries
  - The left term (margin ratio) is greater than the right term
  - Certain types of retail operations may derive a substantial portion of their competitive advantage from selling at a higher margin, rather than higher sales
  - Ex)
- High turnover industries
  - The left term (margin ratio) is lower than the right term
  - Certain types of retail operations may have very low profit margins on sales, and very high turnover, selling a significant multiple of their assets per year
  - Ex)
- The Du Pont analysis allows analysts to determine which of the elements is dominant in any change of ROIC

Margin Ratio

$$\frac{\text{Return}}{\text{Revenue}} = \frac{\text{Revenue} - \text{Fixed costs} - \text{Production Volume} \times \text{Variable costs}}{\text{Revenue}}$$

Production volume ← Flow rate

$$= \frac{\text{Revenue} - \text{Fixed costs} - \text{Flow rate} \times \text{Variable costs}}{\text{Revenue}}$$

Revenue ← Flow rate X Price

$$= 1 - \frac{\text{Fixed costs}}{\text{Flow rate X Price}} - \frac{\text{Variable costs}}{\text{Price}}$$

Flow rate = Min {Input rate, Process Capacity, Demand}

$$= 1 - \frac{\text{Fixed costs}}{\text{Min \{Input rate, Process Capacity, Demand\} X Price}} - \frac{\text{Variable costs}}{\text{Price}}$$

Capital Turnover

$$\frac{\text{Revenue}}{\text{Invested Capital}} = \frac{\text{Flow rate X Price}}{\text{Invested Capital}}$$

Invested Capital = Working capital + PP&E (property, plant and equipment)
= (Current assets – Current liabilities) + PP&E
= Inventory + Account Receivables + Prepayments – Current liabilities + PP&E

$$= \frac{\text{Min \{Input rate, Process Capacity, Demand\} X Price}}{\text{Inventory + Account Receivables + Prepayments - Current liabilities + PP\&E}}$$

Du Pont Analysis - Again

$$\text{ROIC} = \left(1 - \frac{\text{Fixed costs}}{\text{Min \{Input rate, Process Capacity, Demand\} X Price}} - \frac{\text{Variable costs}}{\text{Price}}\right) \times$$

$$\left(\frac{\text{Min \{Input rate, Process Capacity, Demand\} X Price}}{\text{Inventory + Account Receivables + Prepayments} - \underbrace{\text{Current liabilities + PP\&E}}_{\text{Assume uncontrollable}}}\right)$$

ROIC Tree and Simulation

- ROIC
  - $\frac{\text{Return}}{\text{Revenue}} \uparrow$
    - Price ↑
    - Fixed costs ↓
    - Min {Input rate, Process Capacity, Demand} ↑
    - Variable Costs ↓
  - $\frac{\text{Revenue}}{\text{Invested Capital}} \uparrow$
    - Min {Input rate, Process Capacity, Demand} ↑
    - Price ↑
    - {Inventory + Account Receivables + Prepayments – Current liabilities + PP&E} ↓

Sample Atomic Activities

- Sample atomic activities
  - Cut wages
  - Change design so that the work required to make a piece of furniture is reduced
  - Reduce the setup times
  - Change the payment terms with the supplier (may not be operation responsibility)
- Discuss how the above sample atomic activities are related to the items in the previous slide
  - Identify more atomic activities and allocate them

Applied Learning

- Distribute a short case and its accompanying podcast if necessary
  - POM 6 (Applied Learning) Bet on Winning DuPont Analysis & Pick 4 Top Stocks
- Students would share their thoughts with peers
- Students should write down their reflections (A reflection sheet)

Thank You